

Test Plan: Water Intake Tracker (watchOS)

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1. Overview

This document defines the test plan for the Water Intake Tracker; a standalone watchOS application built in SwiftUI. The app lets a user set a personalized daily hydration goal on first launch, log water intake, track progress visually, review a 7-day history, edit their goal at any time, and receive a daily reminder notification. This plan covers onboarding, core functionality, data persistence, edge cases, and notification behavior, along with a risk-based traceability matrix and defect log.

2. Scope

In Scope	first goal onboarding, goal editing, logging functionality, progress ring, undo action, history display, local data saving (UserDefaults), and local notification scheduling.
Out of Scope	iPhone companion app (none exists, this is a standalone Watch app), network/cloud sync (app is fully offline),

3. Test Environment

Device	Apple Watch Series 11 (46mm) Simulator
OS	watchOS 26.5 (Simulator runtime)
IDE	Xcode
Testing Type	Manual, exploratory and scripted functional testing

4. Test Approach & Priority Definitions

Test cases are assigned a priority.

Priority	Level	Description
P1	Critical	Failure blocks app usage (e.g., broken onboarding, lost data)
P2	High	Failure significantly degrades the experience but does not block usage (e.g., undo not working, missing initial prompt)
P3	Medium	Issues with low user impact (e.g., history row ordering)

5. Onboarding & Goal Setup Test Cases

Covers the first-launch goal setup flow and the ability to edit the goal afterward.

ID	Test Case	Steps	Expected Result	Priority	Status
TC-18	First launch shows Goal Setup screen	1. Fresh install (or reset simulator). 2. Launch app for the first time.	Goal Setup screen appears before any other screen. Main screen is not reachable yet.	P1 - Critical	Pass
TC-19	Completing setup transitions to main screen	1. On Goal Setup screen, select a goal value. 2. Tap "Get Started."	App transitions immediately to the main screen, showing the selected goal.	P1 - Critical	Pass
TC-20	Onboarding does not reappear on relaunch	1. Complete onboarding once. 2. Force-quit the app. 3. Relaunch.	App goes straight to the main screen. Goal Setup screen does not appear again.	P1 - Critical	Pass
TC-21	Edit Goal reachable from main screen	1. From main screen, tap "Edit Goal."	Edit Goal screen opens.	P2 - High	Pass

ID	Test Case	Steps	Expected Result	Priority	Status
TC-22	Edit Goal pre-fills current goal	1. Set goal to a specific value during onboarding (e.g., 80 oz). 2. Navigate to Edit Goal.	Shows 80 oz already selected, not a default value.	P2 - High	Pass
TC-23	Changing goal updates ring calculation	1. Log water (e.g., 20 oz). 2. Go to Edit Goal, change goal. 3. Save. 4. Return to main screen.	Progress ring and label reflect the new goal immediately, percentage recalculates correctly.	P1 - Critical	Pass
TC-24	Edit Goal Save returns to main screen	1. Open Edit Goal. 2. Change value. 3. Save.	Screen dismisses back to the main screen automatically.	P3 - Medium	Pass
TC-25	Goal persists after app restart	1. Set a custom goal. 2. Force-quit app. 3. Relaunch.	Custom goal is still in effect; does not reset to default.	P1 - Critical	Pass

6. Functional Test Cases

Core logging, progress display, undo, and history behavior.

ID	Test Case	Steps	Expected Result	Priority	Status
TC-01	Log water — single tap	1. Launch app. 2. Tap +8 button once.	Today's total updates to 8 oz. Progress ring fills proportionally.	P1 - Critical	Pass
TC-02	Log water — multiple taps	1. Tap +8. 2. Tap +12. 3. Tap +16.	Today's total updates to 36 oz after each tap, in real time.	P1 - Critical	Pass
TC-03	Progress ring reaches goal	1. Log entries totaling exactly the daily goal.	Ring is fully filled (100%). No overflow or visual glitch.	P2 - High	Pass
TC-04	Progress ring exceeds goal	1. Log entries totaling more than the daily goal.	Ring stays visually capped at 100%. Numeric total still shows true value above goal.	P2 - High	Pass
TC-05	Undo last entry	1. Log one entry (+8). 2. Tap Undo Last.	Entry is removed. Total returns to 0. Undo button disappears.	P2 - High	Pass
TC-06	Undo removes only most recent entry	1. Log +8, then +12. 2. Tap Undo Last.	Only the +12 entry is removed. Total returns to 8, not 0.	P1 - Critical	Pass
TC-07	History reflects today's total	1. Log 20 oz total. 2. Navigate to History.	Today's row shows 20 oz, matching the main screen.	P2 - High	Pass
TC-08	History shows 7 days	1. Navigate to History screen.	Exactly 7 rows are shown, most recent day first.	P3 - Medium	Pass
TC-09	History highlights goal-met days	1. Log a total meeting/exceeding goal. 2. Navigate to History.	Today's row total is displayed in green. Days under goal remain default color.	P3 - Medium	Pass

7. Data Persistence Test Cases

Confirms logged data and goal settings survive app termination and relaunch.

ID	Test Case	Steps	Expected Result	Priority	Status
TC-10	Data persists after app restart	1. Log +8. 2. Force-quit app. 3. Relaunch app.	Today's total still shows 8 oz.	P1 - Critical	Pass
TC-11	Data persists across simulator relaunch	1. Log entries. 2. Stop app from Xcode (Cmd+). 3. Run again (Cmd+R).	Previously logged entries were present and totals are correct.	P1 - Critical	Pass

8. Edge Case Test Cases

Boundary conditions and unusual usage patterns most likely to expose bugs.

ID	Test Case	Steps	Expected Result	Priority	Status
TC-12	Undo with zero entries logged today	1. Launch app fresh (no entries logged). 2. Confirm Undo Last button is not visible.	Undo Last button does not appear when there is nothing to undo. No crash if triggered programmatically.	P2 - High	Pass

ID	Test Case	Steps	Expected Result	Priority	Status
TC-13	Midnight rollover — new day starts at 0	1. Log entries on Day 1. 2. Advance simulator system date to Day 2. 3. Reopen app.	Today's total resets to 0 oz on the new day. Day 1's total is preserved in History.	P1 - Critical	Not yet tested — requires manual date change
TC-14	History with no prior entries	1. Fresh install, no data logged. 2. Navigate to History immediately.	All 7 rows display, each showing 0 oz, with no crash or blank state.	P3 - Medium	Pass

9. Notification Test Cases

Local notification permission and scheduling behavior.

ID	Test Case	Steps	Expected Result	Priority	Status
TC-15	Notification permission prompt appears	1. Fresh install. 2. Launch app for the first time.	System notification permission dialog appears on first launch.	P2 - High	Pass
TC - 16	Daily remainder at scheduled time	1. Grant notification permission. 2. Set simulator time to just before reminder hour. 3. Wait for scheduled time to pass.	A local notification titled "Stay Hydrated" is delivered at the scheduled time	P2 - High	Pass
TC-17	Notification permission denied	1. Deny notification permission when prompted. 2. Continue using the app normally.	App continues to function normally without notifications. No crash or error state.	P2 - High	Pass

10. Defect Log

Template for tracking defects found during future test passes. No defects were identified during this pass; this table is included as the format that will be used during future expansion.

Defect ID	Description	Severity	Status	Notes / Root Cause
—	No defects found in this test pass.	—	—	—

11. Known Limitations / Not Yet Tested

One case, TC-13 requires manipulating the simulator's system clock to fully verify and was not tested in this pass. It is P1 priority and should be revisited.

No automated testing exists yet for this project; all test cases above were executed manually.

12. Summary

25 total test cases defined across five categories (onboarding/goal setup, functional, persistence, edge case, notification). 24 of 25 passed manual verification; 1 is pending further testing due to constraints (system clock manipulation). No defects were found during this test pass. All P1-priority test cases have been executed at least once, apart from the date-rollover case (TC-13), which remains the highest-priority open item for the next test cycle.